Experimental Configuration	
Recommendation users (sampled)	30,000
Catalog tracks	19,054
User–track interactions	216,170
Interaction sparsity	0.9996
Sampled negatives per user	100
Collaborative model	Truncated SVD
SVD latent components	64
Audio features	13 (MinMax + L2)
Semantic encoder	TF–IDF (1–2 grams)
TF–IDF max features	3,000
Evaluation protocol	Leave–one–out
Ranking cut-offs (K)	5, 10, 20
Random seed	42